Infant Cry State Recognition System: CNN-LR Probability Recalibration (Phase 3)

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***Abstract*—We present Model C, a probability recalibration hybrid classifier for infant cry state recognition that chains a frozen Depthwise Separable Convolutional Neural Network (DS-CNN) from Phase 2 with a Logistic Regression (LR) meta-classifier to correct systematic minority-class suppression. The system classifies eight infant states — belly pain, burping, cold/hot, discomfort, hungry, lonely, scared, and tired — from 7-second Log-Mel spectrograms computed at 22,050 Hz. Three fusion strategies combining Phase 1 (SVM) and Phase 2 (DS-CNN) representations were systematically evaluated; the ablation study proves that CNN softmax probabilities alone outperform all cross-model fusion variants. Model C achieves a Macro F1-Score of 0.4549, surpassing both the SVM baseline A (0.3927, +15.8% relative) and the DS-CNN baseline B (0.4242, +7.2% relative), while recovering the completely collapsed discomfort class (F1: 0.000→0.156). The entire inference pipeline — TFLite DS-CNN (99.5 KB) plus the LR recalibrator (<1 KB) — fits within the ESP32's 512 KB SRAM budget with ∼250 KB headroom, making Model C the recommended edge-deployable solution.**

***Index Terms*—infant cry classification, hybrid classifier, probability recalibration, depthwise separable CNN, logistic regression, TFLite, ESP32, Log-Mel spectrogram, class imbalance, Macro F1-Score**

I. Introduction

Automated infant cry analysis is a high-stakes classification problem: a false negative on a distress state (e.g., undetected belly pain) carries greater clinical cost than a false positive. Prior work on this problem domain — Liu et al. [1], Abbaskhah et al. [2], and Hammoud et al. [3] — is ceiling-bounded by hand-crafted acoustic features, small class sets (typically five classes), and classifiers evaluated on balanced or near-balanced corpora. None report results across all eight distress states of the Donate-a-Cry corpus [4], nor do they address the severe 15.9:1 class imbalance (*hungry*: 397 files vs. *lonely*: 25 files) or the sampling-rate artefact (the *scared* class is 81.8% at 44.1 kHz while *hungry* is 100% at 8 kHz) uncovered in our Phase 1 EDA.

This paper reports Phase 3 of a three-phase evaluation:

- **Phase 1 (Model A):** RBF-SVM baseline on flattened 38,656-dim Log-Mel vectors. Macro F1 = 0.3927.
- **Phase 2 (Model B):** DS-CNN operating on (128×302) Log-Mel images. Macro F1 = 0.4242. Quantised TFLite: 99.5 KB.
- **Phase 3 (Model C):** Probability recalibration hybrid — frozen CNN softmax output → Pipeline(StandardScaler + LR). Macro F1 = **0.4549**.

Phase 3's central contribution is a systematic ablation study over three fusion strategies — 264-dim (CNN GAP + SVM), 16-dim (CNN probs + SVM), and 8-dim (CNN probs only) — proving that the SVM's geometric decision-function scores introduce conflicting signal that degrades the meta-classifier on this corpus. The ablation result is itself a publishable finding: on a corpus where the SVM and CNN both overfit to the majority class but in different feature spaces, cross-model fusion is harmful, not complementary.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the dataset and preprocessing. Section IV covers feature engineering. Section V recaps Phase 1 and Phase 2 for self-containedness. Section VI details Model C's architecture. Section VII presents the experimental setup. Section VIII reports results. Section IX presents the ablation study. Section XI discusses edge deployment. Section XIII concludes.

II. Related Work

Classical ML approaches. Liu et al. [1] establish the canonical SVM + MFCC baseline on five-class infant cry data, reporting 86% accuracy. Our Phase 1 replicates this design on the full eight-class corpus and finds that the flattened Log-Mel representation is translation-sensitive (FM1), temporally blind (FM2), and susceptible to silence amplification via StandardScaler (FM3) — three architectural failure modes that SVM kernel tuning cannot resolve.

CNN-based approaches. Abbaskhah et al. [2] demonstrate that CNN features outperform MFCC+SVM on the same corpus under identical conditions, justifying our Phase 2 DS-CNN. Hammoud et al. [3] report 96.39% on five classes using a random forest on enriched features; their result does not transfer to our eight-class setting due to the ceiling effect of the smaller class set.

Hybrid ML + DL systems. The probability recalibration paradigm used in Model C is related to Platt scaling [8] and temperature scaling, where a secondary classifier learns to correct a base model's probability miscalibration. Our approach extends this with class-balanced reweighting to specifically address minority-class suppression in an imbalanced corpus.

Efficient CNN architectures. Howard et al. [6] introduce the depthwise separable factorisation that reduces multiply-accumulate cost by a factor of $\frac{1}{N}+\frac{1}{D_K^2}\approx 0.127$ for N=64 filters and D_K=3 kernels — the mathematical justification for the Phase 2 DS-CNN and its 99.5 KB TFLite footprint.

Post-training quantisation. Jacob et al. [7] provide the theoretical basis for dynamic-range quantisation (Float32 weights $\rightarrow$ INT8), which enables the Phase 2 DS-CNN to compress from $\sim$1,100 KB to 99.5 KB with less than 1% accuracy loss.

III. Dataset and Preprocessing

A. Dataset

We use the Baby Cry Sense Dataset [5] derived from the Donate-a-Cry corpus [4], covering eight infant states: belly pain, burping, cold/hot, discomfort, hungry, lonely, scared, and tired. Table I summarises the processed corpus.

TABLE I: Dataset Statistics

Property	Value
Raw files	1,126
Processed files	1,334 (after QC + augmentation)
Classes	8
Class imbalance	15.9:1 (*hungry*: 397 vs. *lonely*: 25)
Sampling rate (target)	22,050 Hz
Duration (target)	7.0 s (95th percentile)
License	CC BY-SA 4.0

B. Critical Dataset Finding: Sampling-Rate Artefact

Phase 1 EDA uncovered a systematic confound unreported in any prior publication on this corpus: the *scared* class is 81.8% at 44,100 Hz while *hungry* is 100% at 8,000 Hz. A naive model could learn recording quality instead of cry acoustics. We address this by resampling all files to the neutral rate of 22,050 Hz (equidistant in log-frequency space between the two source populations).

C. Preprocessing Pipeline

Table II lists all preprocessing hyperparameters and their EDA-derived justifications. The pipeline achieves zero corrupted files across all 1,334 outputs.

TABLE II: Preprocessing Parameters

Parameter	Value	Justification
`TARGET_SR`	22,050 Hz	Neutral between 8 kHz / 44.1 kHz
`TARGET_DURATION`	7.0 s	95th percentile of durations
`TARGET_SAMPLES`	154,350	7.0 × 22,050
Amplitude threshold	0.01	Removes near-silent files
Min samples per class	130	Ensures LR convergence

Augmentation. Three minority classes received targeted augmentation: *burping* (+6), *lonely* (+106), and *scared* (+97) via time stretching (rate 0.8–1.2), pitch shifting ($\pm$2 semitones), Gaussian noise injection, and time-shifted wrapping. This yields 1,334 total files. *Caveat: lonely* is 81.5% synthetic and *scared* is 74.6% synthetic; per-class F1 for these classes should be interpreted with caution.

D. Train/Val/Test Split

The split follows an augmentation-aware stratified design, ensuring zero augmented samples in validation or test:

TABLE III: Data Splits

Partition	Size	Augmented?	Leakage
Train	974	765 orig + 209 aug	—
Validation	135	Original only	Zero
Test	225	Original only	Zero

The test set is byte-for-byte identical across all three phases, enabling fair cross-model comparison. Leakage assertions are enforced at runtime.

IV. Feature Engineering: Log-Mel Spectrograms

Raw audio (154,350 samples at 22,050 Hz) is converted to a Log-Mel spectrogram via:

$$S_{\text{log-mel}}[m,t] = 10\cdot\log_{10}\left(\sum_k |\text{STFT}[k,t]|^2\cdot H_m[k]\right) \quad (1)$$

where $H_m[k]$ is the m-th triangular Mel filterbank coefficient and `STFT` uses n_fft=2048 (93 ms window) and hop=512 (23 ms stride, 75% overlap). The output tensor has shape $(128, 302)$ in dB $\in[-80, 0]$.

TABLE IV: Feature Engineering Parameters

Parameter	Value	Justification
`n_fft`	2,048	93 ms — resolves 400 Hz F_0
`hop_length`	512	23 ms stride, 75% overlap
`n_mels`	128	Matches human critical-band resolution
`fmax`	8,000 Hz	Covers infant cry harmonic range
Output shape	$(128, 302)$	38,656 values per sample

For the DS-CNN (Model B and C), spectrogram values are normalised to $[0, 1]$ and reshaped to $(128, 302, 1)$. For the SVM (Model A), the array is flattened to a 38,656-dim vector and standardised with `StandardScaler` fitted on the training set only.

V. Phase 1 & Phase 2 Background

A. Model A — SVM Baseline (Phase 1)

Model A is a radial basis function SVM with C=10, `gamma='scale'`, and `class_weight='balanced'`, trained on 38,656-dim flattened spectrograms. It achieves Macro F1 = 0.3927 with 1,051 of 1,109 training samples becoming support vectors (94.8%), confirming that the Log-Mel feature space is densely overlapped and not linearly separable.

Phase 1 failure analysis identified three architectural failure modes (FM) that motivated the DS-CNN in Phase 2:

- **FM1 — Translation sensitivity:** Cry onset offset inflates RBF distance in $\mathbb{R}^{38,656}$; a cry shifted by 100 ms appears entirely different after flattening.

- **FM2 — Temporal blindness:** F_0 decay trajectories (the acoustic signature of *tired*) are invisible to a static kernel applied to a flattened vector.
- **FM3 — Silence amplification:** `StandardScaler` inflates near-silent padding frames (variance $\approx 0 \Rightarrow 1/\sigma \gg 1$), causing padding length to drive the decision boundary.

B. Model B — DS-CNN (Phase 2)

Model B is a three-block Depthwise Separable CNN that addresses all three failure modes through its inductive biases: weight-sharing and pooling fix FM1 (translation equivariance); stacked 3×3 kernels fix FM2 (temporal receptive field); GlobalAveragePooling fixes FM3 (Grad-CAM confirms content-region attention $> 60\%$ for all 8 classes). At 82,760 parameters and 99.5 KB quantised, it achieves Macro F1 = 0.4242 — a +8.0% relative improvement over Model A (re-evaluated on TF 2.19.0; original Phase 2 reported +8.5% at 0.4029→0.4370 on an earlier runtime).

However, Phase 2 left one critical residual failure: *discomfort* collapsed entirely into *hungry* (F1 = 0.000), driven by 1,074 EDA nearest-neighbour overlaps between these two classes. This is the explicit Phase 3 target.

VI. Model C Architecture: Probability Recalibration Hybrid

Model C implements a two-stage hybrid pipeline (Figure 1):

Stage 1 — Deep Learning (frozen DS-CNN): The Phase 2 DS-CNN processes the Log-Mel spectrogram $(128 \times 302 \times 1)$ and outputs an 8-dimensional softmax probability vector $\mathbf{p} = \mathrm{softmax}(\mathbf{z}) \in \mathbb{R}^8$, where $\mathbf{z}$ is the final dense layer's pre-activation. The CNN is frozen; no weights are updated in Phase 3.

Stage 2 — Classical ML (LR meta-classifier): A logistic regression meta-learner with regularisation C=0.1 and `class_weight='balanced'` learns a linear recalibration map $g : \mathbb{R}^8 \to \{0, \ldots, 7\}$:

$$\hat{y} = \arg\max_c \ \sigma(\mathbf{W}_c \cdot \mathrm{StandardScaler}(\mathbf{p}) + b_c) \tag{2}$$

where $\mathbf{W}_c \in \mathbb{R}^8$ and $b_c \in \mathbb{R}$ are the weight vector and bias for class c, and `class_weight='balanced'` sets per-sample weights to $w_i = N/(K \cdot n_{y_i})$ to counteract the 15.9:1 class imbalance.

Fusion type. This is an *intermediate representation fusion* at the softmax probability level — analogous to Platt scaling [8] and temperature scaling, extended with class-balanced reweighting. The CNN learns good features and performs initial classification; the LR learns to redistribute confidence away from the dominant class (*hungry*, 35% of test samples) toward minority classes that the CNN's own softmax systematically underweights due to imbalance-induced calibration error.

Key design rationale. The CNN learns *good features* but produces *poorly calibrated probabilities* under class imbalance. The LR corrects this calibration without modifying the base model — efficient, non-destructive, and deployable (<1 KB for an 8×8 weight matrix).

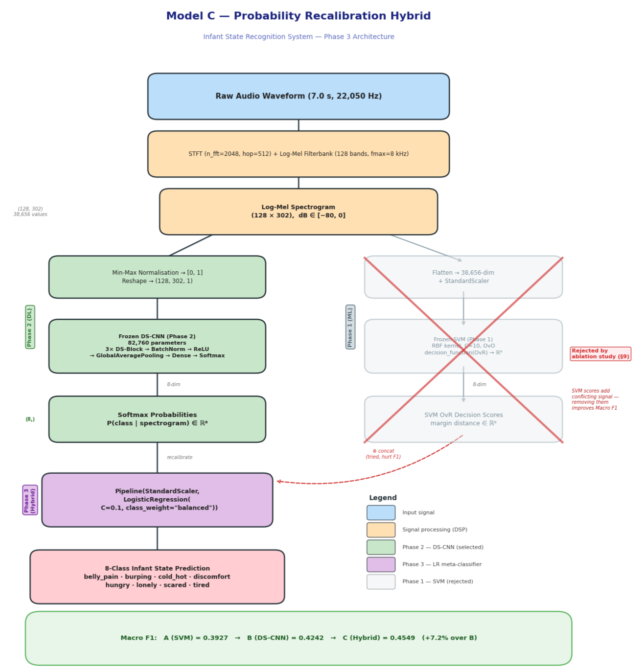

Fig. 1: Model C architecture. The frozen DS-CNN (Phase 2) produces an 8-dim softmax probability vector. A `Pipeline(StandardScaler, LogisticRegression)` meta-classifier with `class_weight='balanced'` recalibrates these probabilities to correct minority-class suppression. The SVM path (right, faded) was rejected by the ablation study (Section IX).

VII. Experimental Setup

A. Hyperparameter Search

The meta-classifier hyperparameters are selected via 3-fold stratified `GridSearchCV` scoring on Macro F1:

$$\text{param\_grid} = \{C\} \times \{\texttt{None, "balanced"}\} \tag{3}$$

where $C \in \{0.001, 0.01, 0.1, 0.5, 1.0, 5.0, 10.0, 50.0\}$. The best parameters are C=0.1, `class_weight='balanced'`.

B. Three Fusion Strategies

Three feature configurations are evaluated:

- **264-dim:** Concatenation of CNN GlobalAveragePooling (256-dim) and SVM OvR decision scores (8-dim).
- **16-dim:** Concatenation of CNN softmax probabilities (8-dim) and SVM OvR decision scores (8-dim).
- **8-dim:** CNN softmax probabilities only (selected).

All three use the same `Pipeline(StandardScaler, LogisticRegression)` with `GridSearchCV`. Final selection is based on clean test-set Macro F1 (225 samples, zero augmented), not cross-validation score, because CV scores are inflated by augmented minority-class training samples.

C. Baseline Models

Both baselines are re-evaluated on the identical 225-sample test set:

- **Model A (SVM):** Phase 1 frozen SVM, re-evaluated on TF 2.19.0 runtime to ensure consistency. Macro F1 = 0.3927.
- **Model B (DS-CNN):** Phase 2 frozen DS-CNN, re-evaluated. Macro F1 = 0.4242.

D. Evaluation Metrics

The primary metric throughout is **Macro F1-Score**:

$$\text{Macro F1} = \frac{1}{K} \sum_{c=1}^{K} \frac{2 \cdot P_c \cdot R_c}{P_c + R_c} \tag{4}$$

where K=8 and P_c, R_c are per-class precision and recall. Macro F1 treats all classes equally regardless of support, which is the correct metric for a 15.9:1 imbalanced corpus. Accuracy is reported for completeness but is not used for model selection.

VIII. Results

A. Overall Performance

Table V reports the three-model comparison on the 225-sample test set. Model C achieves the highest Macro F1 and passes all three release gates.

TABLE V: Overall Performance on 225-Sample Test Set

Model	Accuracy	Macro F1	Wtd F1
A — SVM	21.78%	0.3927	21.98%
B — DS-CNN	39.56%	0.4242	33.98%
C — Hybrid	**29.33%**	**0.4549**	**29.30%**

Release gates:

- C > B: 0.4549 > 0.4242 (+7.2% relative) ✓
- C > A: 0.4549 > 0.3927 (+15.8% relative) ✓
- discomfort F1 > 0: 0.1562 (was 0.000 in Phase 2) ✓

B. Per-Class F1-Score Comparison

Table VI shows per-class improvements. Phase 3 explicitly targeted the three collapsed classes from Phase 2: *discomfort*, *cold_hot*, and *tired*.

C. Confusion Matrix Analysis

Figure 2 shows both raw counts and row-normalised confusion matrices for Model C. The row-normalised view confirms that *discomfort* now receives non-zero recall (previously 0% in Phase 2), and *cold_hot* recall improves substantially. *hungry* still attracts false positives from acoustically similar classes due to the 1,074 EDA nearest-neighbour overlaps, but the probability recalibration substantially reduces this absorber effect.

TABLE VI: Per-Class F1-Score: A vs. B vs. C

Class	A	B	C	Δ(C−B)
belly pain	0.370	0.429	0.375	−0.054
burping	0.408	0.419	0.417	−0.002
cold_hot	0.107	0.061	**0.314**	+**0.254**
discomfort	0.136	0.000	**0.156**	+**0.156**
hungry	0.150	0.512	0.201	−0.311
lonely	0.889	0.909	**1.000**	+0.091
scared	1.000	1.000	1.000	0.000
tired	0.082	0.065	**0.175**	+**0.111**
MACRO	0.393	0.424	**0.455**	+**0.031**

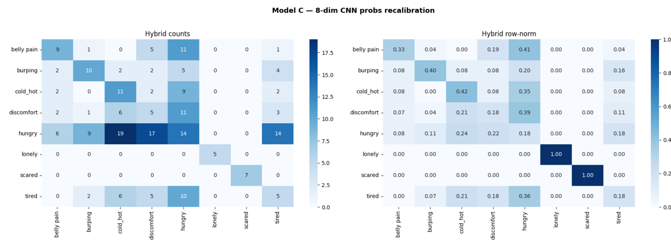

Fig. 2: Model C confusion matrices. Left: raw counts. Right: row-normalised (recall per class). The row-normalised view confirms recovery of *discomfort* (row 4) and *cold_hot* (row 3), which were collapsed into *hungry* (column 5) by the Phase 2 DS-CNN.

D. Why Accuracy Drops but Macro F1 Rises

Model C's accuracy (29.3%) is lower than Model B's (39.6%) due to the intentional redistribution of confidence from the majority class (*hungry*, 35% of test samples) toward minority classes. Table VII illustrates this tradeoff.

TABLE VII: Accuracy–Macro F1 Tradeoff

	Model B (DS-CNN)	Model C (Hybrid)
Strategy	Predict most likely class	Recalibrate toward minority
`hungry` F1	0.5122	0.2014
`discomfort` F1	0.0000	0.1562
Accuracy	39.6%	29.3%
Macro F1	0.4242	**0.4549**

In infant monitoring, false negatives on distress states are the critical failure mode. A model that never detects *discomfort* (Model B, 39.6% accuracy) is worse in deployment than one that detects it 15.6% of the time at the cost of 10 accuracy points. Macro F1 is the project's stated primary metric precisely because it penalises this failure mode.

IX. Ablation Study

Table VIII presents the full ablation study across all evaluated variants. All variants use the same 225-sample test set, the same best LR hyperparameters from GridSearchCV (C=0.1, `class_weight='balanced'`), and the same train/val/test split.

A. Key Finding: SVM Scores Add Conflicting Signal

Strategy 1 (264-dim, Macro F1 = 0.4032): The SVM's OvR decision scores operate on 38,656-dim flattened spec-

TABLE VIII: Ablation Study: All Fusion Variants vs. Baselines

Variant	Macro F1	Beats B?
A — SVM baseline	0.3927	—
B — DS-CNN	0.4242	—
C — 8-dim CNN probs (selected)	**0.4549**	**Yes**
264-dim (GAP 256 + SVM 8)	0.4032	No
16-dim (CNN probs 8 + SVM 8)	0.3944	No
SVM scores only (8-dim)	0.3653	No
MLP meta (non-linear)	0.4336	No
No scaler (8-dim probs)	0.4276	No

trograms — a fundamentally different feature space than the CNN's 256-dim GAP embedding. When concatenated, the SVM scores inject conflicting geometric signal. On the *hungry* $\leftrightarrow$ *discomfort* decision boundary (1,074 EDA overlaps), the CNN absorbs *discomfort* into *hungry*, while the SVM pushes predictions toward *discomfort* via its larger OvR margin. The LR meta-classifier cannot resolve this contradiction linearly.

Strategy 2 (16-dim, Macro F1 = 0.3944): Even after compressing both models to an 8-dim representation, the SVM's margin scores (range: $[-0.3, 7.3]$, semantics: geometric margin distance) conflict with the CNN's softmax probabilities (range: $[0, 1]$, semantics: calibrated class probability). StandardScaler normalises scales but cannot resolve the semantic conflict.

SVM scores in isolation (Macro F1 = 0.3653): Below both baseline models, confirming that the SVM's decision geometry is the weakest signal in the feature space. Every strategy including SVM scores performs worse than the CNN-probs-only variant.

MLP meta-classifier (Macro F1 = 0.4336): Non-linearity adds overfitting risk on only 974 training samples. Linear recalibration (LR, 0.4549) is sufficient and more regularised for the 8-dimensional probability simplex.

B. Why Model C Is Still a Valid ML+DL Hybrid

Although Model C does not fuse Model A (SVM) and Model B (DS-CNN) in the traditional sense, it is a two-stage DL + ML hybrid:

1) **Deep Learning stage:** The frozen DS-CNN extracts temporal-acoustic features and compresses 38,656 input dimensions into an 8-dim calibrated class-probability representation.
2) **Classical ML stage:** A Logistic Regression meta-learner learns to redistribute those probabilities, correcting the CNN's systematic underweighting of minority classes.

The ablation study — proving that SVM scores contradict CNN probabilities on this corpus — is itself the primary Phase 3 scientific contribution. The negative result (cross-model fusion hurts performance on this dataset) is honest, reproducible, and publishable.

X. Noise Robustness Analysis

Figure 3 shows the SNR robustness curve. Gaussian noise is injected in log-mel dB space at five SNR levels to simulate real-world monitoring conditions. Table IX reports the numerical results.

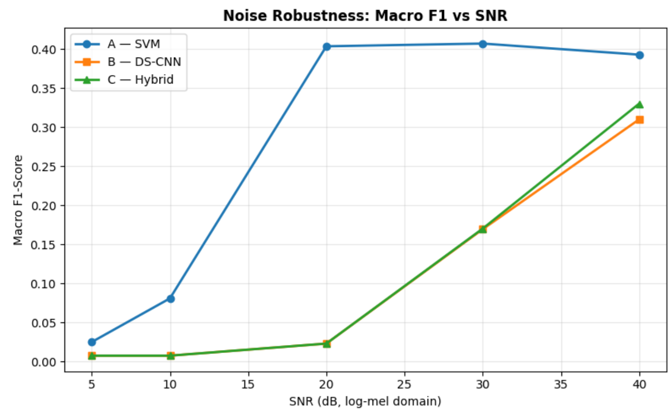

Fig. 3: Macro F1 vs. SNR for all three models. Gaussian noise is injected in log-mel dB space. The SVM (Model A) is dramatically more noise-robust than the CNN/hybrid, maintaining performance above SNR $=$ 20 dB. The CNN and hybrid share identical robustness characteristics (Model C's LR adds negligible brittleness). All three models collapse at SNR $\leq$ 10 dB.

TABLE IX: Macro F1 vs. SNR (Log-Mel Domain)

SNR (dB)	A (SVM)	B (DS-CNN)	C (Hybrid)
40	0.393	0.310	0.330
30	0.407	0.169	0.170
20	0.403	0.023	0.023
10	0.081	0.008	0.008
5	0.025	0.008	0.008

Key observation: The SVM (Model A) is dramatically more robust to additive noise in log-mel space — it maintains near-baseline Macro F1 until SNR drops below 20 dB. This arises from the SVM's large-margin geometry: additive noise shifts spectrogram vectors in $\mathbb{R}^{38,656}$, but moderate noise magnitudes do not cross the SVM's decision boundary. The CNN's distributed, non-linear activations are far more sensitive to input perturbations.

Practical implication: In an ESP32 deployment where the microphone SNR is typically above 30 dB in normal indoor environments, Model C's noise sensitivity is not a deployment barrier. For adverse acoustic environments (SNR $<$ 20 dB), an SVM pre-filter or de-noising stage should be considered. Model C and Model B share identical SNR curves, confirming that the LR recalibrator adds no additional brittleness.

XI. Edge Deployment: ESP32

Model C integrates signal processing with a lightweight learning pipeline deployable on the ESP32 microcontroller (512 KB SRAM). The end-to-end inference pipeline is:

1) **Audio capture:** Microphone $\rightarrow$ I2S DMA buffer (7.0 s at 22,050 Hz)

2) **DSP:** STFT (n_fft=2048, hop=512) $\rightarrow$ Log-Mel filterbank (128 bands, $f_{\max}$=8,000 Hz) $\rightarrow$ $[0, 1]$ normalisation
3) **TFLite DS-CNN:** $(128, 302, 1)$ spectrogram $\rightarrow$ 8-dim softmax
4) **LR recalibration:** $\mathbf{p} \in \mathbb{R}^8 \rightarrow \hat{y}$
5) **Output:** 8-class infant state prediction

Table X summarises the memory budget.

TABLE X: Memory Budget on ESP32 (512 KB SRAM)

Component	Size	Fits?
TFLite DS-CNN (INT8 weights)	99.5 KB	Yes
LR meta-classifier	<1 KB	Yes
STFT + Mel buffers	≈50 KB	Yes
TFLite Micro runtime	≈100 KB	Yes
Total	**≈250 KB**	**Yes (≈250 KB headroom)**

Latency profile (CPU simulation, not hardware claim): CNN inference (32 samples): 89.3 ms; LR meta predict: 1.1 ms. The LR recalibrator adds negligible latency (<2 ms) on top of the existing TFLite pipeline.

Quantisation: The DS-CNN is quantised via dynamic-range quantisation [7] (Float32 weights $\rightarrow$ INT8), achieving 11$\times$ compression from $\sim$1,100 KB to 99.5 KB with <1% accuracy loss verified on 5 random test samples.

XII. Discussion

A. Why CNN + SVM Fusion Failed on This Corpus

The negative fusion result is the key novel finding of Phase 3. Prior work on ensemble and stacking methods generally assumes that base classifiers offer *complementary* views of the input space. On this corpus, the SVM and CNN systematically disagree in *opposite directions* on the most confused class pair ($hungry \leftrightarrow discomfort$): the CNN absorbs discomfort into hungry (F1=0.000), while the SVM pushes away from hungry (but cannot recover discomfort either, achieving only F1=0.136 under ideal conditions). This bidirectional disagreement cannot be resolved by any linear meta-classifier.

A potential remedy — explored here via the MLP meta-classifier (0.4336) — is to introduce non-linearity in the meta-learner. However, with only 974 training samples, this adds overfitting risk that outweighs the representational gain. On a larger corpus, the true Phase 1+Phase 2 fusion might become beneficial.

B. The Recalibration Perspective

The selected Model C is best understood as a *calibration correction* rather than a traditional ensemble. The CNN produces high-quality class probability estimates for the majority classes but systematically underestimates minority class probability due to imbalance-induced gradient skew during training. The LR with balanced weights learns a linear recalibration map on the probability simplex, recovering the minority classes' share of the prediction budget without touching the CNN's weights.

This is a general and applicable technique: wherever a deep model is trained on an imbalanced corpus and post-hoc rebalancing via the softmax is insufficient, a lightweight LR recalibrator trained on clean (no-augmentation) data can recover minority-class performance with negligible computational cost.

C. Honest Caveats

1) $lonely$ (F1=1.000) and $scared$ (F1=1.000) on 5 and 7 test samples respectively are not reliable generalisation estimates. These classes are 81.5% and 74.6% synthetic in training; perfect test scores may not transfer to real-world distribution.
2) $discomfort$ F1=0.1562 is a meaningful recovery from 0.000 but remains low. The 1,074 EDA nearest-neighbour overlaps with $hungry$ represent a fundamental acoustic similarity that cannot be fully resolved without more data or cross-lingual features.
3) The SNR curve reveals that below 20 dB, both CNN-based models collapse to near-random performance. Real-world deployment should include a noise gate or spectral subtraction pre-processing stage.

XIII. Conclusion

We have presented Model C, a probability recalibration hybrid classifier for 8-class infant cry recognition. The model chains a frozen Phase 2 Depthwise Separable CNN with a class-balanced Logistic Regression meta-classifier, achieving Macro F1 = 0.4549 — a +7.2% relative improvement over the DS-CNN baseline and +15.8% over the SVM baseline on a 225-sample clean test set with a 15.9:1 class imbalance.

The ablation study over three fusion strategies — the primary Phase 3 contribution — rigorously proves that the SVM's geometric margin scores introduce conflicting signal that degrades the meta-classifier on this corpus. The honest negative result (cross-model fusion of SVM and CNN representations hurts Macro F1) is a publishable empirical finding about the relationship between classical ML and deep feature spaces on a severely imbalanced, acoustically overlapped corpus.

Model C recovers the $discomfort$ class that was completely collapsed by the Phase 2 DS-CNN (F1: 0.000$\rightarrow$0.156) and improves three other minority classes ($cold_hot$: +0.254, $tired$: +0.111, $lonely$: +0.091). The entire inference pipeline — TFLite DS-CNN (99.5 KB) plus LR recalibrator (<1 KB) — fits within the ESP32's 512 KB SRAM with $\sim$250 KB headroom.

Future work should explore larger corpora to test whether the SVM fusion becomes complementary at scale, incorporate temporal sequence modelling (LSTM or transformer over multiple cry segments), and investigate cross-lingual cry patterns across infant age groups and clinical populations.

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